A PROJECT REPORT

**ON**

# ACCIDENT HOTSPOT PREDICTION USING RANDOM FOREST AND SPATIAL CLUSTERING

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING**
**BACHELOR OF TECHNOLOGY**
**In**
**A.Y.: 2026-27**

**By**


**Gadiga Cheran Tej Reddy – 2520030466**

**Athineni Arun Tej – 2520030472**

**Earla Ashwith Reddy – 2520030475**
**Ravindra Gaddam**
**Professor, Department of CSE**


## Department of Computer Science and Engineering


**Koneru Lakshmaiah Education Foundation**
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

## Abstract

< This project presents the design, implementation, and evaluation of a machine learning system for [problem/domain]. The study covers data collection, preprocessing, exploratory analysis, feature engineering and/or selection, model development, training, testing, hyperparameter tuning, and evaluation. Appropriate machine learning algorithms are selected according to the problem characteristics. Experimental results are compared using suitable evaluation metrics, followed by interpretation of model performance, limitations, and practical implications.>

Keywords: Accident hotspots, Random Forest, spatial clustering, K-Means, road safety, risk prediction

# 1. Introduction

## 1.1 Purpose of the Project

Road accidents create loss of life, injuries, traffic disruption, and economic impact. In a city, accident risk is not uniform: particular road stretches, junctions, and corridors may repeatedly record higher accident and fatality counts. Identifying these locations from historical records can help authorities prioritise safety actions.

The main stakeholders are traffic police, transport authorities, urban planners, road-safety teams, and city administrators. A machine-learning based approach can summarise historical patterns and provide a repeatable way to classify locations according to accident risk. This project focuses on the supplied Hyderabad City black-spot records for 2023–2025.

## 1.2 Problem Statement

The project treats accident-risk identification as a multi-class classification problem. For each black-spot location, historical accident and fatality information is transformed into numerical features and a severity score. Locations are assigned LOW, MEDIUM, or HIGH risk using the documented severity rule, and a Random Forest classifier predicts the risk class. In parallel, K-Means groups locations with similar accident-risk profiles to support hotspot analysis.

## 1.3 Existing Methods and Disadvantages

Traditional hotspot analysis commonly relies on accident counts, manually prepared black-spot lists, simple thresholds, or descriptive statistics. These approaches are useful for reporting but may not capture interactions among multiple accident and fatality features. GIS-based methods can provide strong spatial analysis when reliable geographic coordinates are available.

Machine-learning methods can model nonlinear relationships and combine multiple predictors, but their performance depends on data quality, feature selection, class balance, and validation. The supplied dataset does not contain latitude/longitude, so true coordinate-based clustering cannot be performed directly. This project therefore uses feature-space clustering as an explicit analytical proxy and documents that limitation.

## 1.4 Proposed System

The proposed pipeline loads the supplied Hyderabad City black-spot CSV, removes the four banner/header rows, standardises text fields, converts numerical fields, handles missing numerical values with zero, removes

duplicates, recalculates yearly totals, and engineers average yearly accidents, accident growth, fatality rate, severity score, and stretch length. Categorical zone information is one-hot encoded. A stratified 70:30 train/test split is used with random_state=42, followed by a Random Forest classifier with 100 trees and maximum depth 5. K-Means with three clusters is applied to standardised accident, fatality, severity, and average-yearly-accident features.

### 1.5 Research Objectives( atleast 4)

1. Prepare and validate the Hyderabad City black-spot dataset for analysis.
2. Clean the dataset and perform exploratory data analysis of accidents and fatalities.
3. Engineer accident-risk features and prepare a leakage-aware feature matrix.
4. Train a Random Forest classifier and perform feature-space K-Means clustering.
5. Evaluate the classifier using accuracy, precision, recall, F1-score, and a confusion matrix.
6. Interpret the results and identify high-risk locations for practical traffic-safety prioritisation.

## 2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

| S.No. | Title | Reference with DOI | Problem or gap addressed | Objective of the Research | Focus of the paper (Methods used) | Inference | Recommendations for future research |
|---|---|---|---|---|---|---|---|
| 1 | A Machine Learning Approach for Classifying Road Accident Hotspots | Paiva et al., 2023, ISPRS IJGI, 12(6), 227, DOI 10.3390/ijgi12060227 | Hotspot classification needs multiple road and spatial factors. | Compare classifiers for severe/non-severe hotspot risk. | SVM, Random Forest, MLP | ML can classify hotspot risk; MLP performed best. | Test additional cities and richer spatial data. |
| 2 | Prediction of Traffic Accident Severity Based on Random Forest | Yang et al., 2023, Journal of Advanced Transportation, 2023, 7641472, DOI 10.1155/2023/7641472 | Traditional severity models may miss nonlinear interactions. | Develop RF severity model and rank features. | Random Forest and feature importance | RF can rank important accident features and improve severity prediction. | Use broader datasets and additional factors. |
| 3 | Evaluating the effectiveness of machine learning techniques in forecasting the severity of traffic | Obasi & Benson, 2023, Heliyon, 9(8), e18812, DOI 10.1016/j.heliyon.2023.e18812 | Accident severity is influenced by many interacting factors. | Compare several ML classifiers. | Naive Bayes, RF, Logistic Regression, ANN | RF and Logistic Regression achieved strong accuracy on UK data. | Extend to richer casualty information. |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | accidents | | | | | | |
| 4 | Accident severity prediction modeling for road safety using random forest algorithm: an analysis of Indian highways | Khanum et al., 2023, F1000Research, 12, 494, DOI 10.12688/f1000research.133594.2 | Indian highway severity prediction can suffer from imbalance and limited variables. | Build and tune RF for Indian highway severity. | RF, feature selection, tuning | Model performance varied strongly between training and test data. | Improve data balance and validation. |
| 5 | Traffic accident severity prediction with ensemble learning methods | 2024, Computers and Electrical Engineering, 114, 109101, DOI 10.1016/j.compeleceng.2024.109101 | Need robust multi-class severity classification. | Compare ensemble classifiers. | RF and AdaBoost | RF achieved high classification performance on urban accident data. | Test external datasets and additional ensemble models. |
| 6 | Utilizing GIS and Machine Learning for Traffic Accident Prediction in Urban Environment | Khan & Hussain, 2024, Civil Engineering Journal, 10(6), 1922–1935, DOI 10.28991/CEJ-2024-010-06-013 | Urban hotspot identification benefits from GIS and ML together. | Identify hotspots and predict accident occurrence. | Moran's I, RF, Decision Tree, Linear Regression | GIS hotspot analysis and ML complement each other. | Add richer road and traffic features. |
| 7 | Comparison of road crash severity prediction using tree-based machine learning algorithms with varying data treatments | Acharya & Henry, 2024, JSCEIIAI, 5(1), 15–24, DOI 10.11532/jsceiiai.5.1_15 | Data treatment can change tree-model performance. | Compare tree-based models under different treatments. | Decision trees and Random Forest | Preprocessing choices affect crash-severity prediction. | Evaluate more treatment strategies. |
| 8 | Analysis of Road Traffic Crash | Garg et al., 2024, EASTS, 15, 3228–3244, DOI 10.11175/easts.15.3228 | Indian highway crash severity is | Analyse severity and contributi | ML, hierarchical clustering | Clustering and ML can reveal severity | Integrate richer road and environme |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Severity Through Machine Learning on Indian Highways | | affected by complex factors. | ng factors. | , feature importance | patterns. | ntal variables. |
| 9 | Revealing crash hotspots concerning google traffic maps historical data by supervised and ensemble machine learning techniques | Mirzahossein et al., 2025, Transportation Engineering, 20, 100326, DOI 10.1016/j.treng.2025.100326 | Static historical data may miss changing traffic conditions. | Predict crash hotspots using dynamic traffic information. | XGBoost and one-class SVM | Dynamic traffic information can improve hotspot prediction. | Integrate real-time traffic and weather data. |
| 10 | Road Traffic Accident Hotspot Detection: A GIS-Based Machine Learning Approach Using HDBSCAN and Spatial Clustering Techniques | Roy et al., 2026, Geographies, 6(2), 55, DOI 10.3390/geographies6020055 | Hotspot patterns vary over time and space. | Combine GIS, clustering, and temporal hotspot analysis. | HDBSCAN, LISA-Moran's I, scan statistics | Multiple clustering approaches reveal evolving hotspot zones. | Use higher-resolution geocoded datasets. |

# 3. Methodology

## 3.1 Proposed System Flow Diagram

Insert the proposed-system flow diagram here as perf your project.

Recommended flow: Dataset → Data Cleaning → Missing-Value Treatment → Encoding → Feature Engineering/Selection → Train/Test Split → Model Training → Hyperparameter Tuning → Prediction → Evaluation → Final Model.

## 3.2 Dataset Description

The dataset is the supplied Hyderabad City black-spot list identified by the state for calendar year 2025, containing historical accident and fatality counts for 2023, 2024, and 2025. It contains 49 black-spot locations and

15 raw fields after the source headers are removed. The raw fields include serial number, zone, police station, road information, location name, start/end kilometre points, yearly accident counts, yearly fatality counts, and three-year totals. The preprocessing stage derives additional features including average yearly accidents, accident growth, fatality rate, severity score, and stretch length. The target variable is the derived risk_level with 20 LOW, 17 MEDIUM, and 12 HIGH locations.

## 3.3 Experimental Setup

Experimental setup: standard laptop or desktop PC suitable for Python-based machine learning; Windows 11; Python 3.10; Visual Studio Code. The implementation uses pandas, NumPy, scikit-learn, matplotlib, and seaborn. The project requirements specify pandas >=2.0.0, NumPy >=1.24.0, scikit-learn >=1.2.0, matplotlib >=3.7.0, and seaborn >=0.12.0.

## 3.4 Data Preprocessing

- Data inspection and schema validation were performed after removing the four source banner/header rows.
- Duplicate records were removed.
- Missing numerical values were converted to numeric values and filled with zero.
- Extreme values were retained because they represent real high-accident locations.
- Zone categories were one-hot encoded.
- StandardScaler was used for K-Means clustering features.
- Risk labels were generated from the documented severity-score thresholds.
- A stratified 70:30 train/test split with random_state=42 was used; the target label was excluded from the input matrix.

## 3.5 Exploratory Data Analysis

[Include distributions, class balance, correlation analysis, scatter plots, box plots, and other relevant visualizations.]

## 3.6 Feature Engineering and Feature Selection (if applicable)

Derived features were average_yearly_accidents, accident_growth, fatality_rate, severity_score, and stretch_length_km. The Random Forest input matrix used yearly accident counts, yearly fatality counts, the derived numerical features, and one-hot encoded zone categories. The target risk_level was excluded from the predictors to avoid direct target leakage.

## 3.7 Model Building and Training

The main predictive model is RandomForestClassifier from scikit-learn. The implementation uses 100 decision trees, maximum depth 5, and random_state=42. The dataset is split into 34 training records and 15 test records using a stratified 70:30 split. The classifier learns the three risk classes and produces class predictions for the test set.

## 3.8 Hyperparameter Tuning(if applicable)

The supplied implementation does not run GridSearchCV or RandomizedSearchCV. The Random Forest parameters are explicitly set in the project code: n_estimators=100, max_depth=5, and random_state=42. Therefore these values are treated as fixed project settings rather than a claim of exhaustive optimisation.

## 3.9 Model Evaluation

Classification evaluation uses accuracy, weighted precision, weighted recall, weighted F1-score, and a confusion matrix. ROC-AUC is not computed by the supplied implementation.

# 4. Results and Discussion

## 4.1 Preprocessing Results

[Report missing values before/after treatment, dimensions, encoding/scaling results, and final feature count.]

## 4.2 Model Performance

**Random Forest: Accuracy 73.33%; weighted Precision 73.33%; weighted Recall 73.33%; weighted F1-score 72.81%. Confusion matrix (Actual × Predicted; LOW, MEDIUM, HIGH): [[5,1,0],[0,4,1],[1,1,2]].**

| Model | Accuracy | Precision | Recall | F1-score | ROC AUC |
|---|---|---|---|---|---|
| Random Forest | 73.33% | 73.33% | 73.33% | 72.81% | Not computed |
| Test set (15 records) | 73.33% | 73.33% | 73.33% | 72.81% | Not computed |
| K-Means clustering | N/A | N/A | N/A | N/A | N/A |
| Project result | 73.33% | 73.33% | 73.33% | 72.81% | Not computed |
| Note | Weighted | Weighted | Weighted | Weighted | Not computed |

## 4.3 Visual Results and discussion

The main visual results for this classification project are the yearly accident/fatality summary, top accident-prone locations, risk-category distribution, and Random Forest feature importance. ROC-AUC is not computed by the project code and is therefore reported as not computed rather than estimated.

The Random Forest model achieved 73.33% accuracy, 73.33% weighted precision, 73.33% weighted recall, and 72.81% weighted F1-score on 15 test records. The confusion matrix shows correct predictions across all three classes, with some confusion between adjacent risk levels. Feature importance indicates that average yearly accidents, accidents in 2023, fatalities in 2023, fatality rate, and accidents in 2025 are among the strongest predictors. Because the dataset is small, the reported performance should be interpreted as a project-level result rather than a production-grade generalisation claim.

# 5. Conclusion

This project developed a machine-learning workflow for accident hotspot and risk analysis using 49 Hyderabad City black-spot locations from 2023–2025. After cleaning and feature engineering, the system created LOW, MEDIUM, and HIGH risk labels from a documented severity score and trained a Random Forest classifier. K-Means clustering was used as a feature-space analytical proxy because geographic coordinates were not present. The Random Forest achieved 73.33% accuracy and 72.81% weighted F1-score on the test set. The resulting analysis can help prioritise locations for traffic-safety review, while further validation with larger and geographically richer data is recommended.

# 6. Future Work

- Use larger, multi-year, and more diverse accident datasets.
- Compare Random Forest with additional ensemble and tree-based algorithms.
- Use grid search or randomized search with cross-validation.
- Add explainable AI methods such as SHAP to interpret model predictions.
- Test robustness on new time periods and monitor model drift.

- Develop a dashboard or deployable application for traffic-safety teams.
- Validate the workflow on other city or state accident datasets containing geographic coordinates.

## 7. References

[Use one consistent style, preferably APA. Include every cited work and verify DOI and bibliographic information.]

[1] Paiva, A. C., et al. (2023). A Machine Learning Approach for Classifying Road Accident Hotspots. ISPRS International Journal of Geo-Information, 12(6), 227. doi: 10.3390/ijgi12060227.

[2] Yang, X., et al. (2023). Prediction of Traffic Accident Severity Based on Random Forest. Journal of Advanced Transportation, 2023, 7641472. doi: 10.1155/2023/7641472.

[3] Obasi, I. C., & Benson, C. (2023). Evaluating the effectiveness of machine learning techniques in forecasting the severity of traffic accidents. Heliyon, 9(8), e18812. doi: 10.1016/j.heliyon.2023.e18812.

[4] Khanum, H., et al. (2023). Accident severity prediction modeling for road safety using random forest algorithm: an analysis of Indian highways. F1000Research, 12, 494. doi: 10.12688/f1000research.133594.2.

[5] (2024). Traffic accident severity prediction with ensemble learning methods. Computers and Electrical Engineering, 114, 109101. doi: 10.1016/j.compeleceng.2024.109101.

[6] Khan, A. A., & Hussain, J. (2024). Utilizing GIS and Machine Learning for Traffic Accident Prediction in Urban Environment. Civil Engineering Journal, 10(6), 1922–1935. doi: 10.28991/CEJ-2024-010-06-013.

[7] Acharya, N., & Henry, M. (2024). Comparison of road crash severity prediction using tree-based machine learning algorithms with varying data treatments. Intelligence, Informatics and Infrastructure, 5(1), 15–24. doi: 10.11532/jsceiiai.5.1_15.

[8] Garg, T., Chatterjee, S., & Maitra, B. (2024). Analysis of Road Traffic Crash Severity Through Machine Learning on Indian Highways. Journal of the Eastern Asia Society for Transportation Studies, 15, 3228–3244. doi: 10.11175/easts.15.3228.

[9] Mirzahossein, H., et al. (2025). Revealing crash hotspots concerning google traffic maps historical data by supervised and ensemble machine learning techniques. Transportation Engineering, 20, 100326. doi: 10.1016/j.treng.2025.100326.

[1] Paiva, A. C., et al. (2023). A Machine Learning Approach for Classifying Road Accident Hotspots. ISPRS International Journal of Geo-Information, 12(6), 227. doi: 10.3390/ijgi12060227.

Signature of the Guide Course Instructor